

EUGENE L. BUCHANAN

Cyber Security Engineer — Offensive & AI-Powered Secure Development



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Professional Summary

Hands-on security engineer with deep application- and product-security experience — white-box testing, integration security, and evidence-based vulnerability discovery — who now runs that same discipline through agentic AI. I ship secure software and verify it: SOC 2 / compliance-driven testing, SAML/OAuth identity and API security, and OWASP-aligned agentic offensive tooling (/bug-reaper, MCP, security-focused agents) covering prompt-injection and AI/LLM red-teaming with a zero-false-positive bias. I turn attacker pathways and compliance requirements into observable, tested, well-bounded controls teams can trust in production — Rust, AWS, and the reliability mindset that makes security verification repeatable, not anecdotal.

Technical Expertise

Application & Web Security Testing	92%	App & Web Security — OWASP-aligned web2: IDOR, auth, SSRF, XSS, SQLi, RCE, API/GraphQL.
Identity & Access Mgmt (SAML, OAuth, OIDC)	90%	Identity & Access — SAML, OAuth, OIDC; deny-by-default principal isolation.
SOC 2 & Compliance (Evidence, Audits)	88%	Compliance (SOC 2) — End-to-end Type II: controls, evidence, audits.
Agentic Offensive Security (bug-reaper, MCP)	90%	Agentic Offensive — LLM agents + MCP; /bug-reaper; AI red-teaming.
White-Box Code Review & Secure SDLC	91%	White-Box SDLC — White-box + code review across backend, API, DB, frontend.
Cloud Security (AWS, IAM, Logging)	84%	Cloud Security — AWS IAM/ECS/S3; Docker, Kubernetes, Terraform.

Technology Stack

- bug-reaperBug BountyPenetration TestingOWASP Top 10IDORAuth BypassSSRFXSSSQLiNoSQLiRCESSTIPrompt InjectionAI Red-TeamingAPI / GraphQLWhite-Box TestingSecure Code ReviewThreat ModelingHackerOneBugcrowd
- Working PoC
- SOC 2 (Type II)SAMLOAuthOIDCSSOSFTPPermission ProfilesPrincipal IsolationDeny-by-DefaultRisk Assessment
- Audit Evidence
- LLM AgentsMCP (Model Context Protocol)JSON-RPCUnix Domain SocketsOpenCodeCodex AgentsHermes AgentRust
- PythonNode.jsReactPostgreSQLAWS (IAM, ECS, S3)Cloud SecurityDockerKubernetesTerraform / IaCGit

Key Strengths

Translations — select a language:

ENESARZHURUDEJA

Translations:

Experience

Security Engineer / Agentic Security Contributor (Open-Source)

2026 – Present

ZeroClaw Labs (github.com/zeroclaw-labs/zeroclaw) · Rust · 32.2k stars

- Run agentic security testing and verification against the codebase using offensive AI skills (/bug-reaper) — evidence-based web2 bug-bounty methodology that finds only real, triage-able bugs: IDOR/access control, auth/session bypass, SSRF, XSS, SQLi/NoSQLi, RCE, and API/GraphQL flaws. 4-phase workflow (recon → audit → validate → report); every finding requires a working PoC with zero false positives.
- Build and operate a security-focused agent stack (LLM agents + MCP tool servers + Unix-domain-socket JSON-RPC) to automate vulnerability verification across the codebase, closing the loop between discovery and verified fix.
- Contributed architectural review of a multi-user auth & isolation PR (62 files, 6,684 additions) — peercred, native pairing, SSH challenge-response, and OIDC providers; deny-by-default permission profiles; principal isolation at RPC dispatch. Mapped the workspace_id multi-tenant axis into the principal abstraction to avoid a second security migration.

Senior Software Engineer / QA Lead / Product Security & Compliance

Dec 2020 – Aug 2025

Knowledgecity LLC · Remote · AWS / Postgres / React

- Owned and drove the SOC 2 Type II filing across security, engineering, and operations — mapped controls (access, change management, logical & physical security, availability, confidentiality, privacy), gathered evidence from AWS / CI/CD / identity systems, closed gaps, and managed the auditor relationship through successful certification.
- Led product/integration security: shipped SAP, Oracle, Workday, Coursera, UKG, and Zoom integrations hardened across SAML, OAuth, SFTP, and custom REST APIs — enforcing identity, session, and API security boundaries across partner auth models and data flows.
- Ran white-box security testing across integration, backend, API, database, and frontend surfaces; built the CI/CD test-and-verification pipeline (Cypress, Selenium, Postman, JUnit, S3) used to prove security and compliance controls continuously.
- Designed security regression suites, health dashboards, and alerting to detect and prove remediation across multi-environment deployments; reduced technical debt behind security controls.
- Mentored and trained the test automation team; ran code reviews with a security-first lens; raised engineering standards.

Senior Software Engineer II / Presales Engineering

RealNetworks · Seattle, WA · Streaming Media Platform

- QA on an advanced research team evaluating distributed system behavior across consumer appliances, mobile platforms, stream servers, and cellular networks — complex, multi-component reliability and protocol scenarios.
- Global presales engineering: turned product capabilities into customer solutions across diverse technical environments; trained new hires; led cross-functional collaboration.

Senior Software Test Engineer IV

Microsoft · Redmond, WA · Windows Media Server

- White-box testing and automation for Windows Media Server, Windows 98/NT — security and verification frameworks for complex system and network-protocol testing.
- Deep debugging across server infrastructure, network protocols, and media codecs.

Education

- Associate Degree, Victor Valley College
- California Notary Commission
- Toastmasters International

Other Interests

Security Research

Reading vulnerability write-ups and open-source exploit tooling; exploring agentic security and MCP-based offense.

Real Estate

Buy-and-hold multifamily and value-add deals, analyzed with a cash-flow-first lens.

Renewable Energy

Solar, storage, and the grid of the future — off-grid/on-grid systems and the engineering behind them.

Tango Dancing

Argentine tango — the embrace, the pause, the musicality of the walk.



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